

Elastic net regularization

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In statistics and, in particular, in the fitting of linear or logistic regression models, the **elastic net** is a regularized regression method that linearly combines the L1 and L2 penalties of the lasso and ridge methods.

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Specification

The elastic net method overcomes the limitations of the LASSO (least absolute shrinkage and selection operator) method which uses a penalty function based on

$$\|\beta\|_1 = \sum_{j=1}^p |\beta_j|.$$

Use of this penalty function has several limitations.^[1] For example, in the "large p , small n " case (high-dimensional data with few examples), the LASSO selects at most n variables before it saturates. Also if there is a group of highly correlated variables, then the LASSO tends to select one variable from a group and ignore the others. To overcome these limitations, the elastic net adds a quadratic part to the penalty ($\|\beta\|^2$), which when used alone is ridge regression (known also as Tikhonov regularization). The estimates from the elastic net method are defined by

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} (\|y - X\beta\|^2 + \lambda_2 \|\beta\|^2 + \lambda_1 \|\beta\|_1).$$

The quadratic penalty term makes the loss function strictly convex, and it therefore has a unique minimum. The elastic net method includes the LASSO and ridge regression: in other words, each of them is a special case where $\lambda_1 = \lambda, \lambda_2 = 0$ or $\lambda_1 = 0, \lambda_2 = \lambda$. Meanwhile, the naive version of elastic net method finds an estimator in a two-stage procedure : first for each fixed λ_2 it finds the ridge regression coefficients, and then does a LASSO type shrinkage. This kind of estimation incurs a double amount of shrinkage, which leads to increased bias and poor predictions. To improve the prediction performance, the authors rescale the coefficients of the naive version of elastic net by multiplying the estimated coefficients by $(1 + \lambda_2)$.^[1]

Reduction to Support Vector Machine

In late 2014, it has been proven that the Elastic Net can be reduced to the linear support vector machine.^[2] A similar reduction has previously been proven for the LASSO in 2014.^[3] The authors show that for every instance of the Elastic Net, an artificial binary classification problem can be constructed such that the hyper-plane solution of a linear support vector machine (SVM) is identical to the solution β (after re-scaling). The reduction immediately enables the use of highly optimized SVM solvers for Elastic Net problems. It also enables the use of GPU acceleration, which is often already used for large-scale SVM solvers.^[4] The reduction is a simple transformation of the original data and regularization constants

$$X \in \mathbb{R}^{n \times p}, y \in \mathbb{R}^n, \lambda_1 \geq 0, \lambda_2 \geq 0$$

into new artificial data instances and a regularization constant that specify a binary classification problem and the SVM regularization constant

$$X_2 \in \mathbb{R}^{2p \times n}, y_2 \in \{-1, 1\}^{2p}, C \geq 0.$$

Here, y_2 consists of binary labels $-1, 1$. When $2p > n$ it is typically faster to solve the linear SVM in the primal, whereas otherwise the dual formulation is faster. The authors refer to the transformation as Support Vector Elastic Net (SVEN), and provide the following Matlab pseudo-code:

```
function beta=SVEN(X,y,t,lambda2);
[n,p]=size(X);
X2 = [bsxfun(@minus, X, y./t); bsxfun(@plus, X, y./t)]';
Y2=[ones(p,1);-ones(p,1)];
if 2p>n then
    w = SVMPrimal(X2, Y2, C = 1/(2*lambda2));
    alpha = C * max(1-Y2.*(X2*w),0);
else
    alpha = SVMDual(X2, Y2, C = 1/(2*lambda2));
end if
beta = t * (alpha(1:p) - alpha(p+1:2p)) / sum(alpha);
```

Software

- "Glmnet: Lasso and elastic-net regularized generalized linear models" is software which is implemented as an R source package.^{[5][6]} This includes fast algorithms for estimation of generalized linear models with ℓ_1 (the lasso), ℓ_2 (ridge regression) and mixtures of the two penalties (the elastic net) using cyclical coordinate descent, computed along a regularization path.
- JMP Pro 11 includes elastic net regularization, using the Generalized Regression personality with Fit Model.
- "pensim: Simulation of high-dimensional data and parallelized repeated penalized regression" implements an alternate, parallelised "2D" tuning method of the ℓ parameters, a method claimed to result in improved prediction accuracy.^{[7][8]}
- scikit-learn includes linear regression, logistic regression and linear support vector machines with elastic net regularization.
- SVEN, a Matlab implementation of Support Vector Elastic Net. This solver reduces the Elastic Net problem to an instance of SVM binary classification and uses a Matlab SVM solver to find the solution. Because SVM is easily parallelizable, the code can be faster than Glmnet on modern hardware.^[9]

References

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2. Zhou, Quan; Chen, Wenlin; Song, Shiji; Gardner, Jacob; Weinberger, Kilian; Chen, Yixin. *A Reduction of the Elastic Net to Support Vector Machines with an Application to GPU Computing*. Association for the Advancement of Artificial Intelligence.
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5. Friedman, Jerome; Trevor Hastie; Rob Tibshirani (2010). "Regularization Paths for Generalized Linear Models via Coordinate Descent". *Journal of Statistical Software*: 1–22.
6. "CRAN - Package glmnet". *r-project.org*.
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8. "CRAN - Package pensim". *r-project.org*.
9. "mlcircus / SVEN — Bitbucket". *bitbucket.org*.

External links

- http://www.stanford.edu/~hastie/TALKS/enet_talk.pdf Regularization and Variable Selection via the Elastic Net (http://www.stanford.edu/~hastie/TALKS/enet_talk.pdf) (presentation)

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